Practical-8

Problem Definition:

The system is designed to generate a sequential lighting pattern using multiple LEDs to create a
"chasing" or running light effect.
Specifically, it performs the following operations:
-> It generates a binary count sequence using flip-flops (0 to 7)
->It uses this sequence to select one output at a time
->Only one LED is turned ON at any given moment
->The LEDs glow in a fixed order (LED0 → LED1 → LED2 → … → LED7)
->The sequence continuously repeats to produce a moving light effect
->The speed of LED transition is controlled using a clock signal
If the system is working correctly, it will create a smooth and continuous LED chasing pattern
where the light appears to move from one LED to the **next.**

Tools/Technology To Be Used:

The following tools and technologies are used to design and implement the LED Chaser Circuit:
Logisim Software – Used for designing and simulating the digital circuit
D Flip-Flops – Used to create a 3-bit counter (0 to 7 sequence)
Clock Generator – Provides timing pulses to control the speed of LED transitions
Multiplexer (8:1 MUX) – Selects one output line based on the counter input
Splitter / Bus – Used to combine and separate multi-bit signals
LEDs – Used to display the output visually
Basic Logic Gates (AND, OR, XOR, NOT) – Used to implement counter logic

Aim:

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"chasing" or running light effect.
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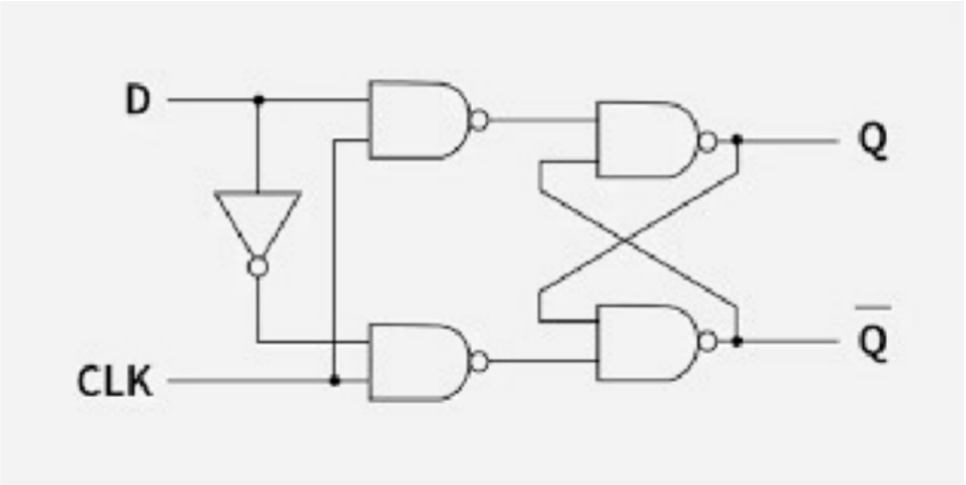
D flip flop:

Truth table:

Q(n)	D(n)	Q(n+1) (Next)
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table:

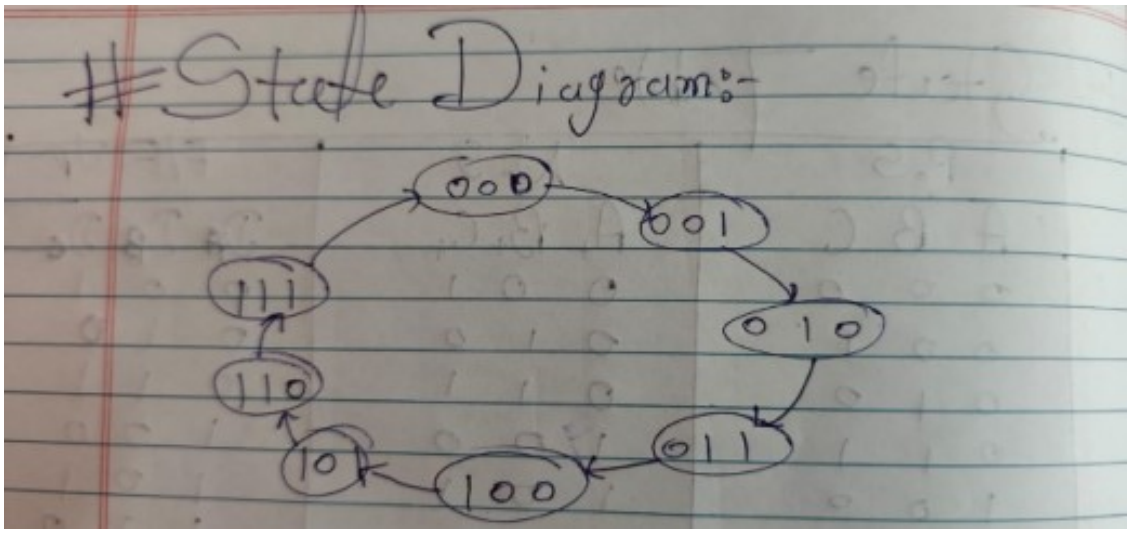
Q Output		Input
$Q_{(n)}$	$Q_{(n+1)}$	D_n
0	0	0
0	1	1
1	0	0
1	1	1

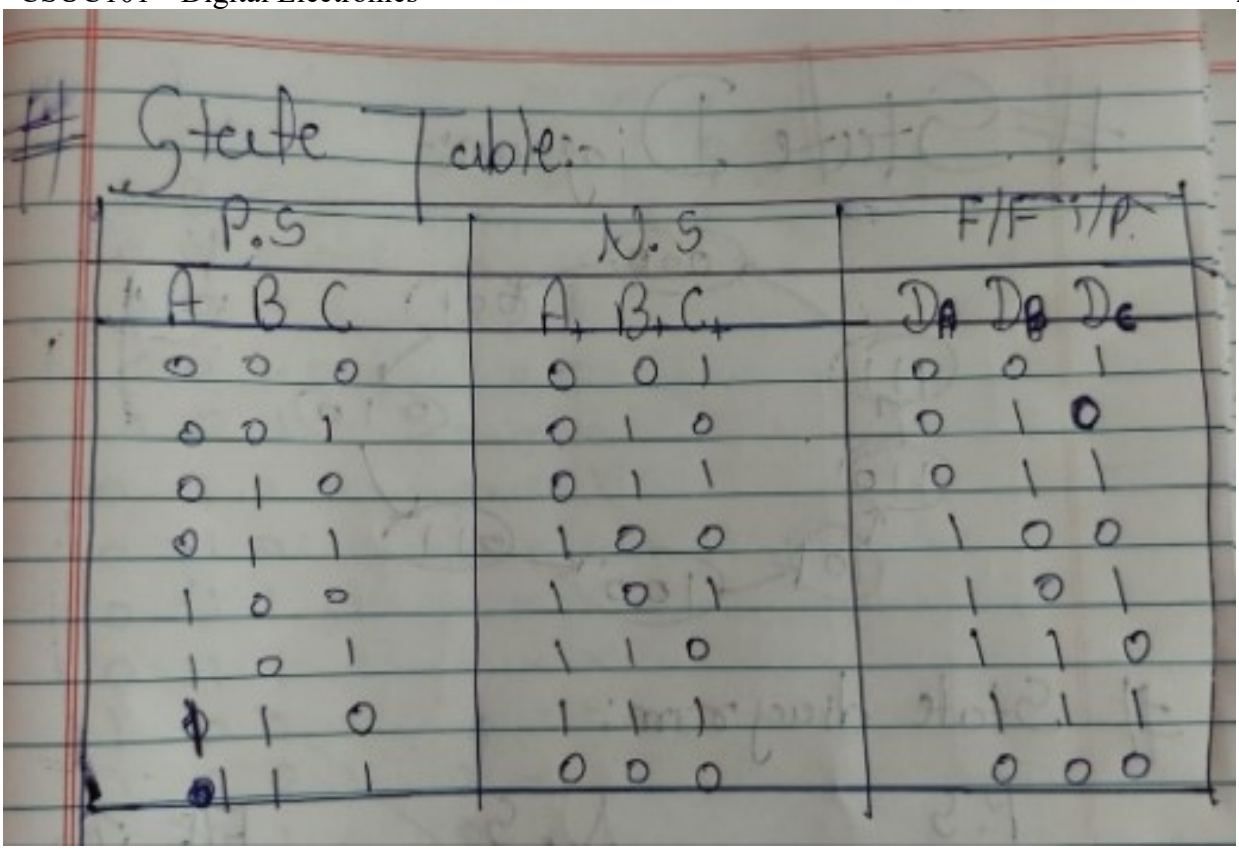


Circuit Diagram:

4-bit binary Random counter:

State diagram & State Table:

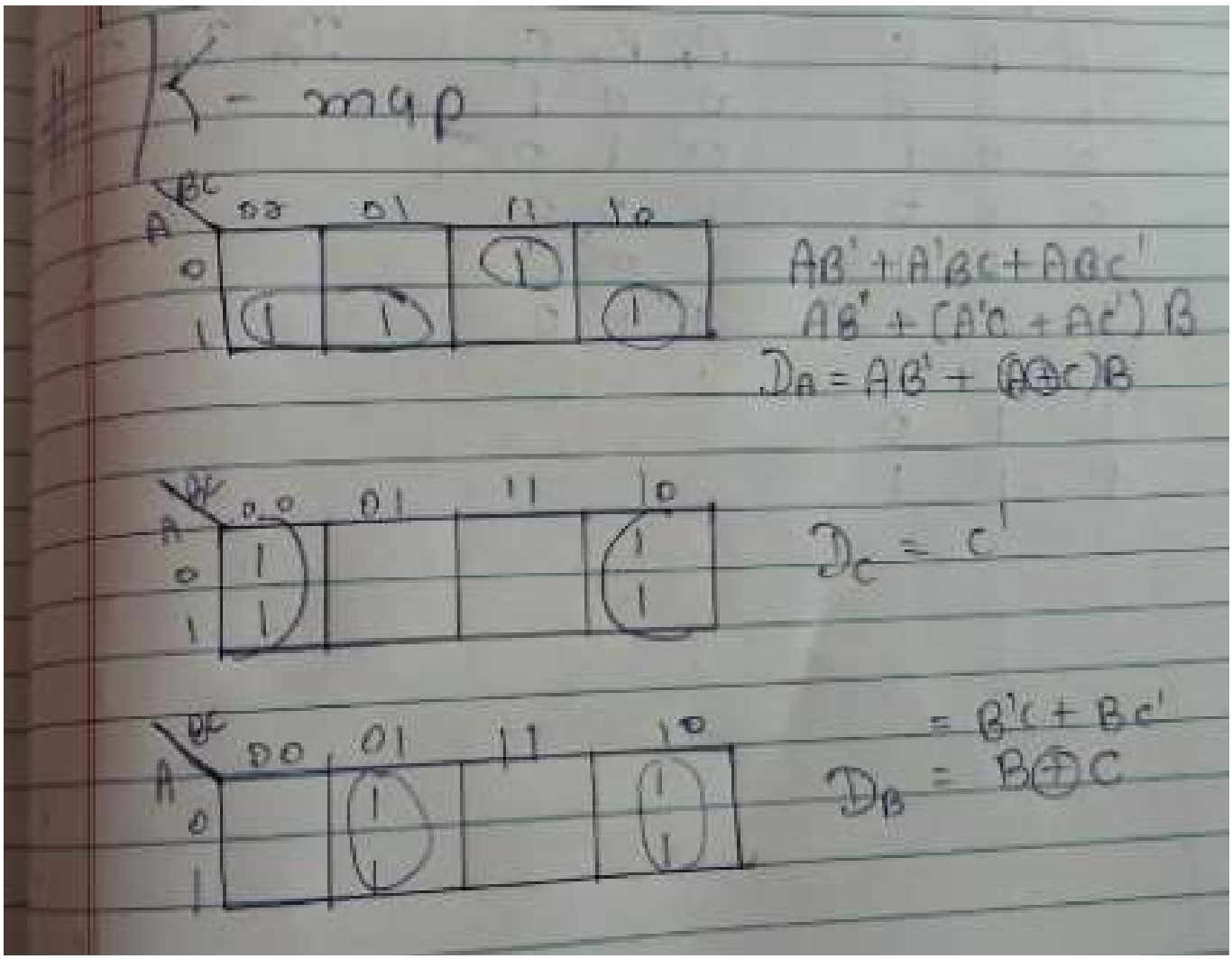




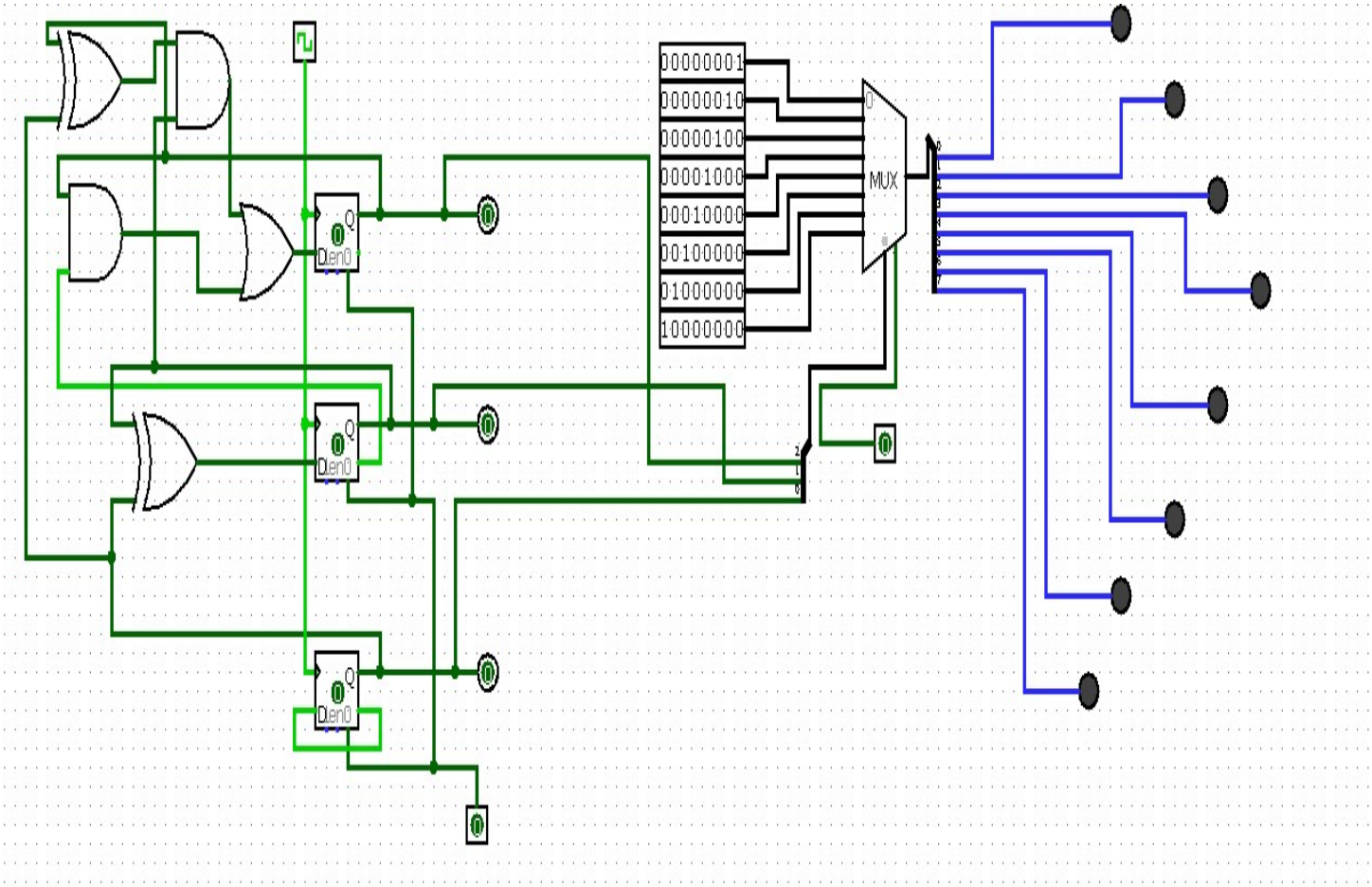
State Table:

P.S			N.S			F/F I/P		
A	B	C	A_+	B_+	C_+	D_A	D_B	D_C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	1	0	1	1	0	1
1	0	1	1	1	0	1	1	0
1	1	0	1	1	1	1	1	1
1	1	1	0	0	0	0	0	0

K-Map & circuit diagram:

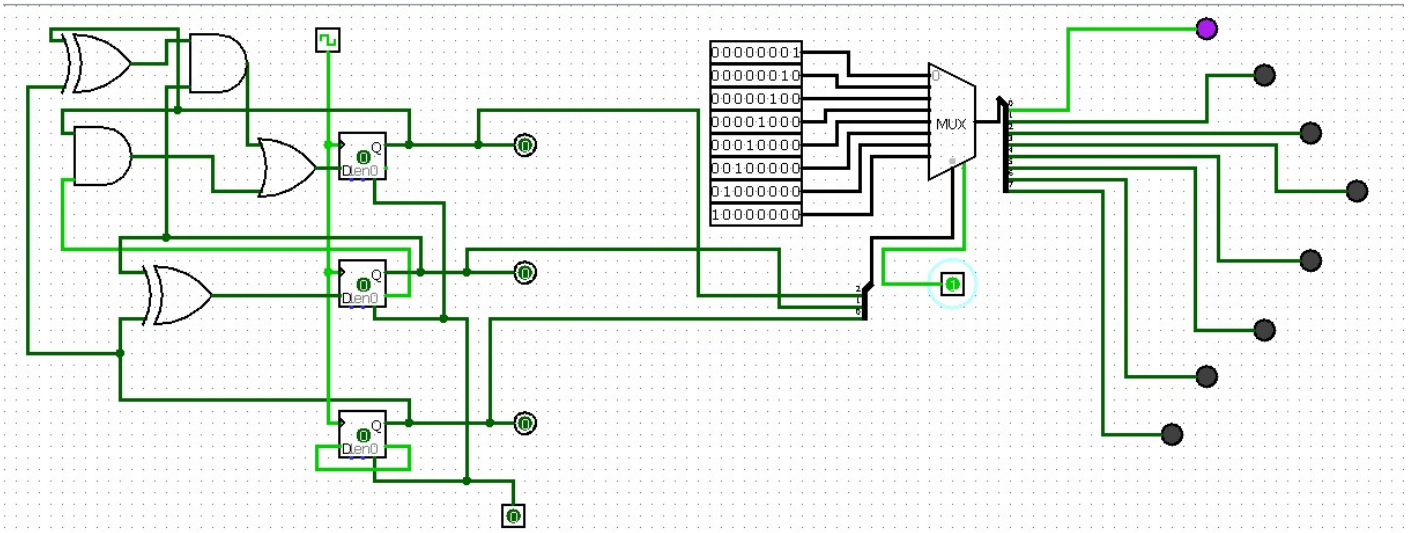


Circuit of Logism :

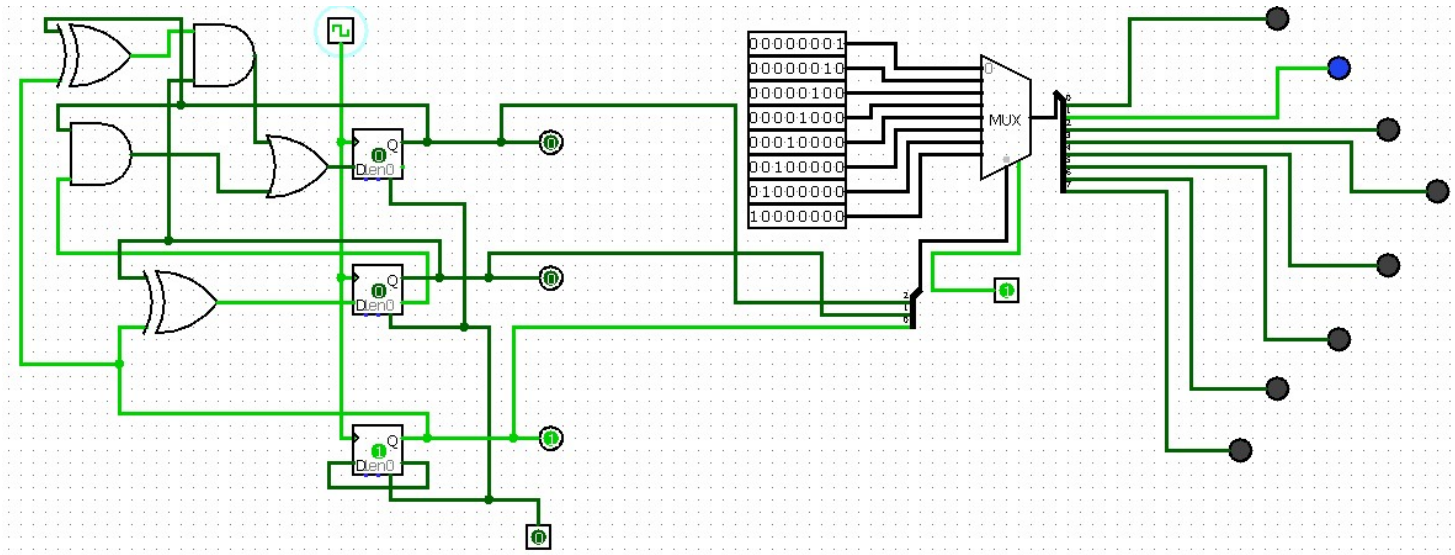


Screen shot:

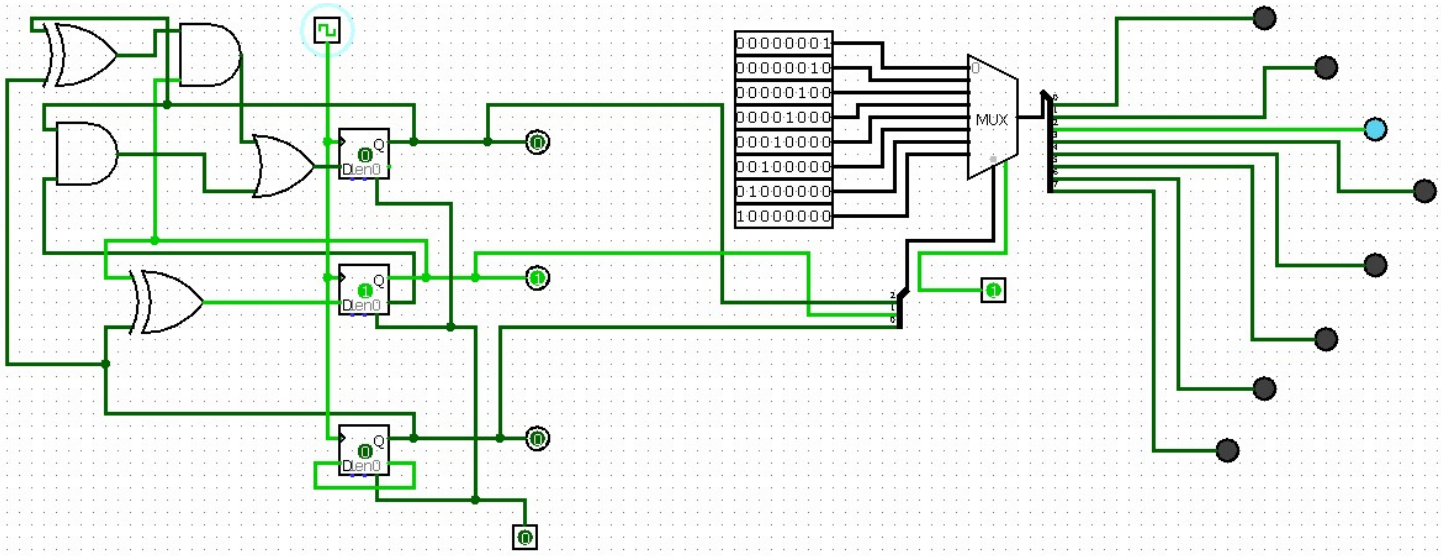
For LED 1:



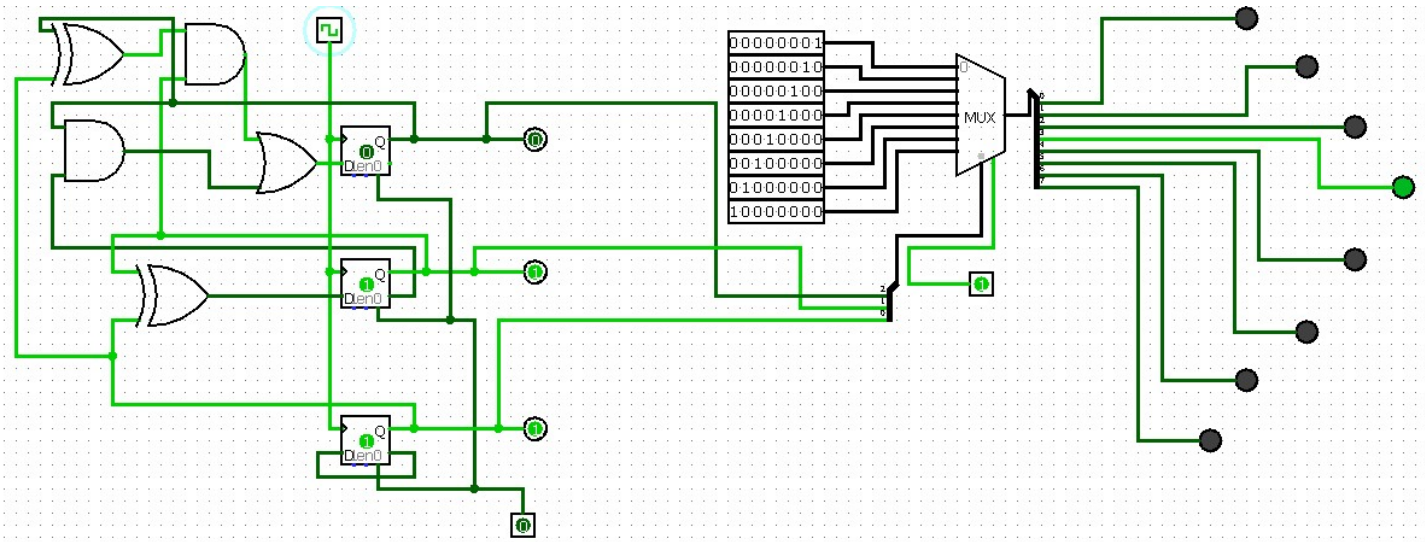
For LED 2:



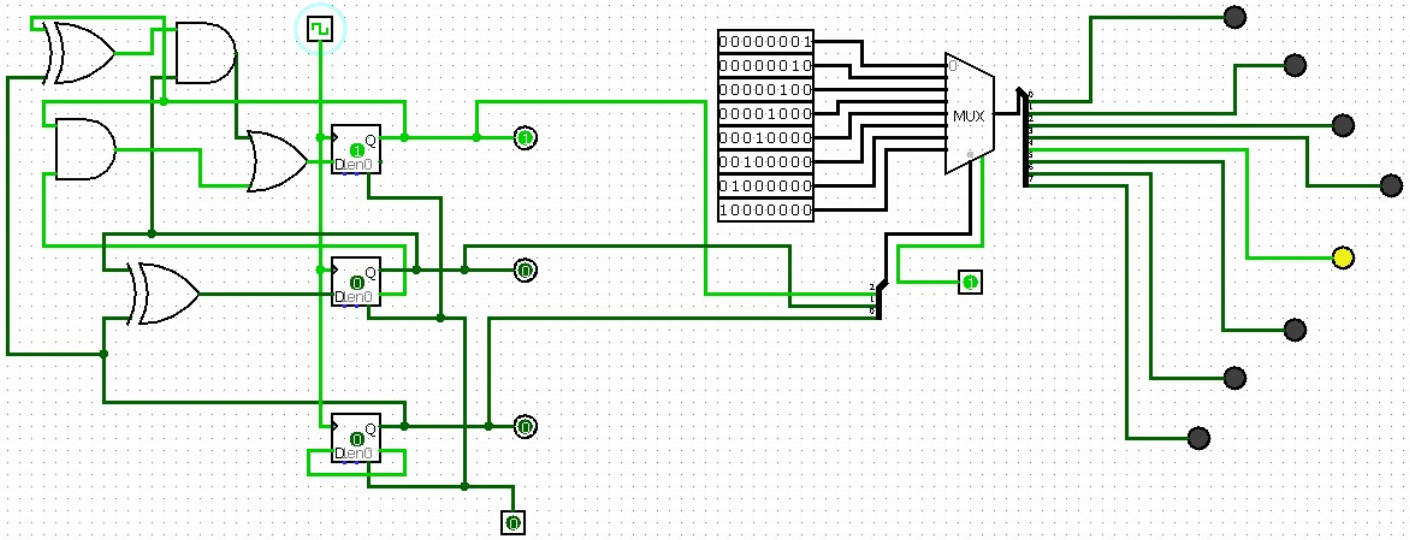
For LED 3:



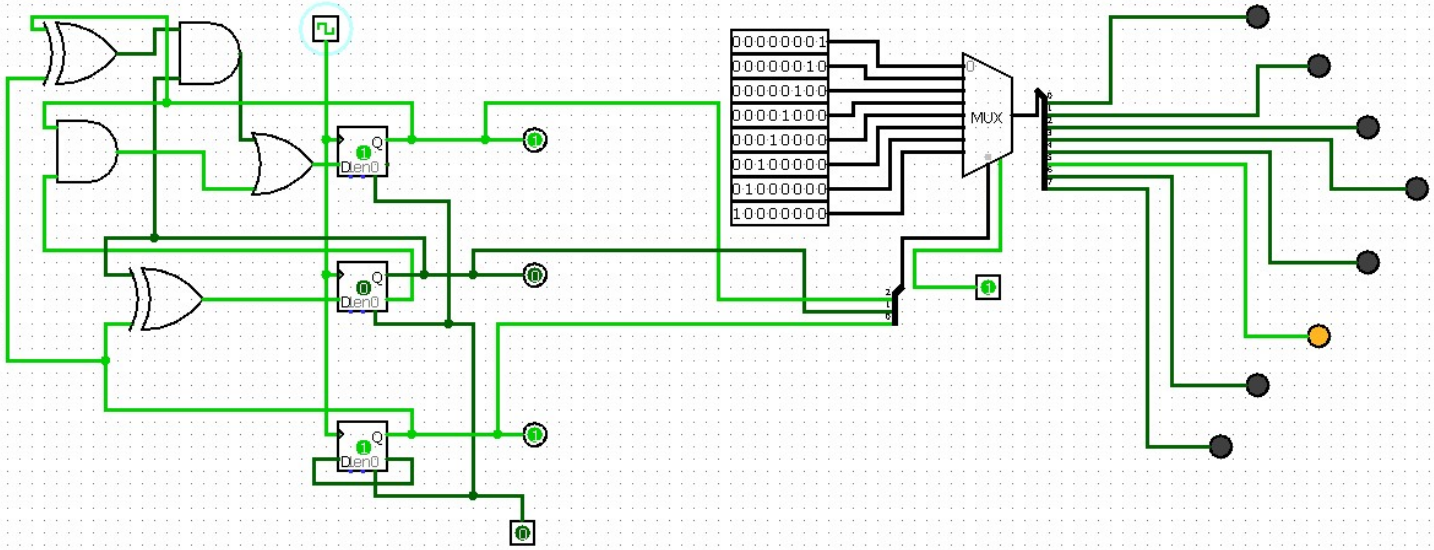
For LED 4:



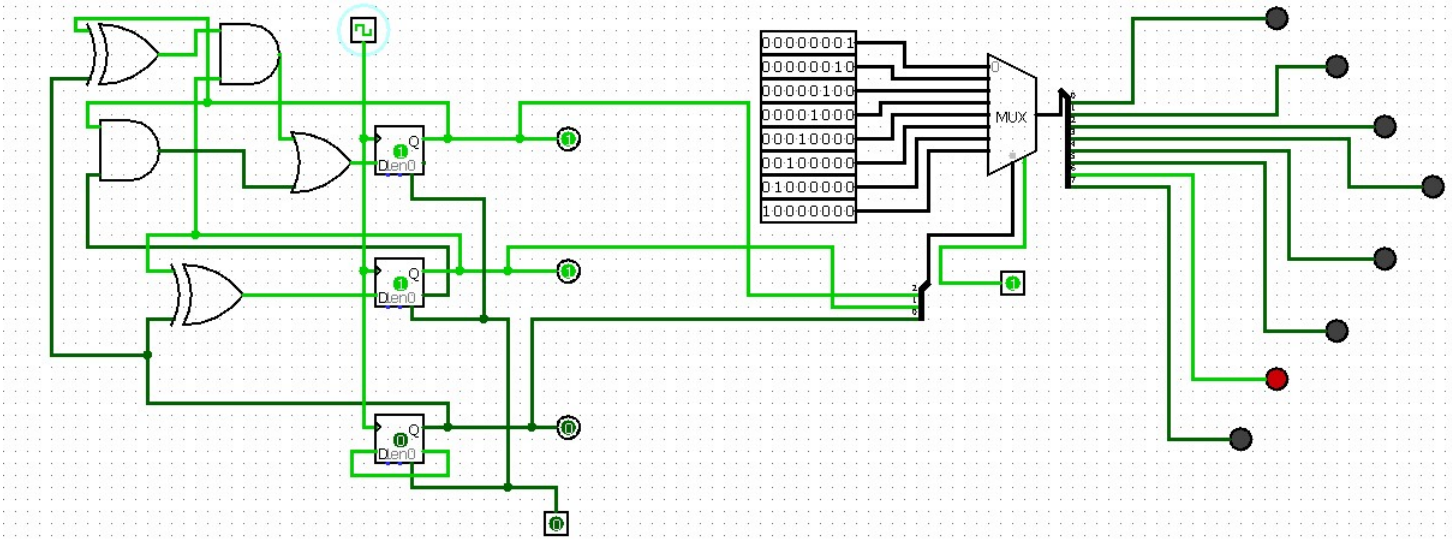
For LED 5:



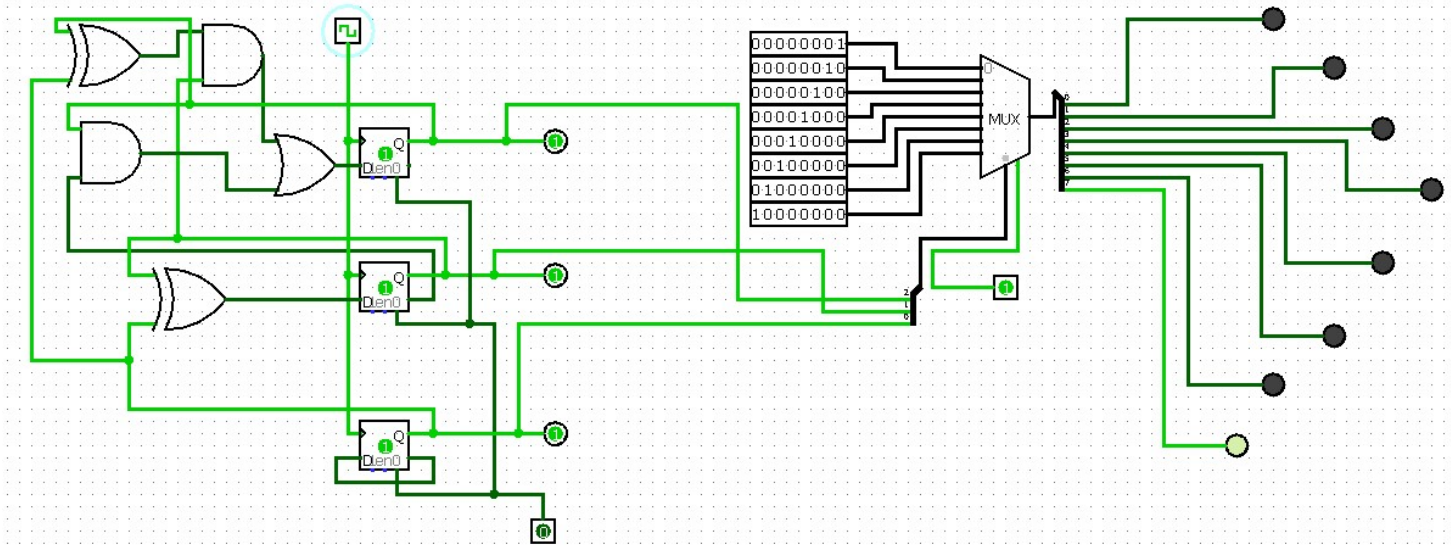
For LED 6:



For LED 7:



For LED 8:



Conclusion : The LED Chaser Circuit was successfully designed and simulated using digital logic components in Logisim. The circuit produced a smooth sequential lighting pattern where LEDs turned ON one at a time using a counter and control logic. This project helped in understanding the practical implementation of sequential circuits, clock signals, and output control in digital systems.